A Federated Digital Twin Framework with Multimodal Data Integration for Personalized Healthcare Monitoring

Thesis Defense

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Thesis Introduction

Introduction

Digital healthcare focuses on continuous monitoring of patient health conditions.Traditional systems rely on centralized data collection, which raises privacy and scalability concerns.

With the increasing use of wearable devices and IoT sensors, healthcare systems must ensure:

- Secure data handling
- Matching
- Real-time health analysis

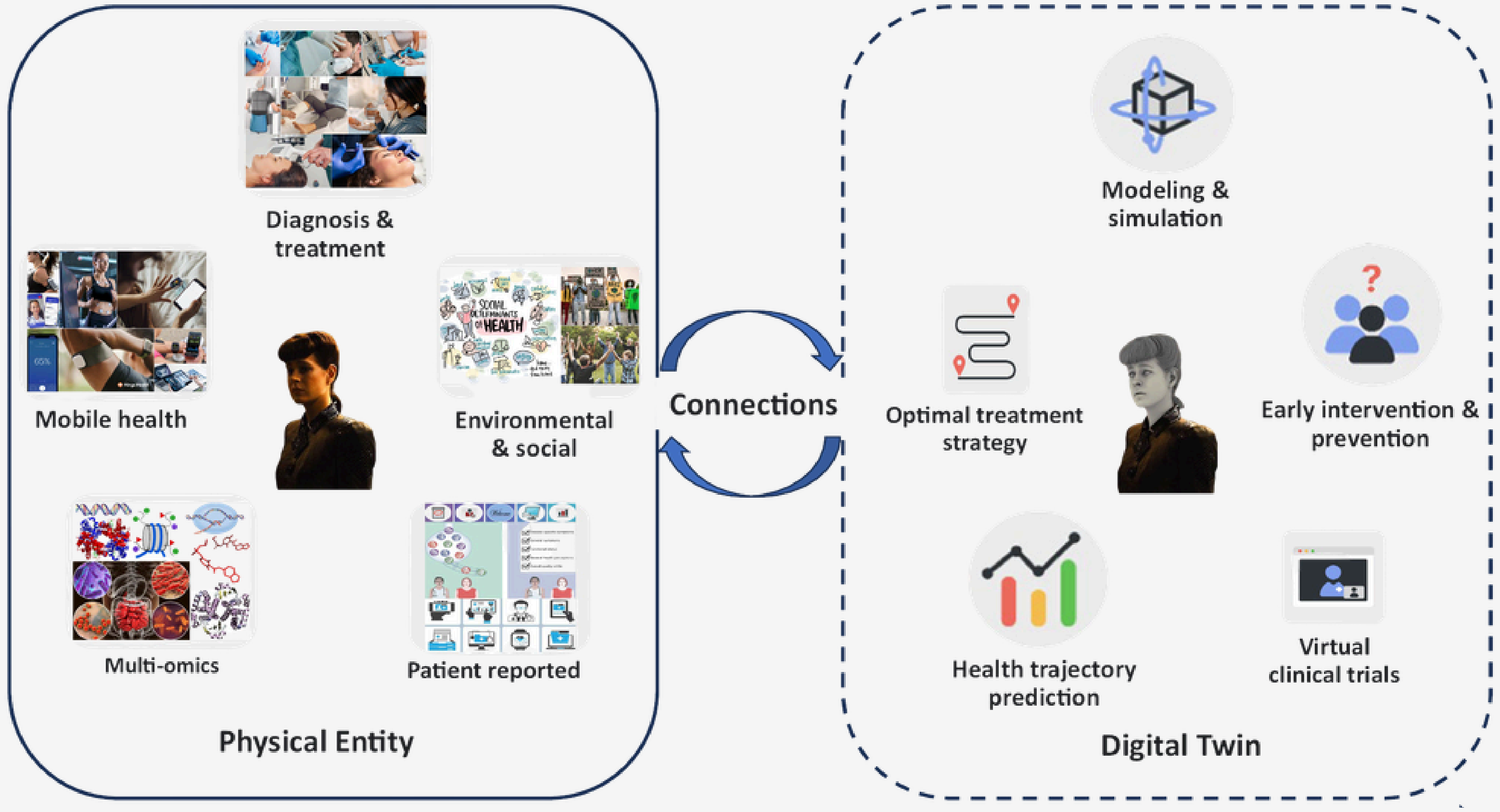

Fig 1:Digital twins for health

Background

- IoT devices and wearable sensors generate large volumes of healthcare data.
- Centralized systems face privacy, security, and regulatory limitations.
- Federated learning supports decentralized and secure data utilization.

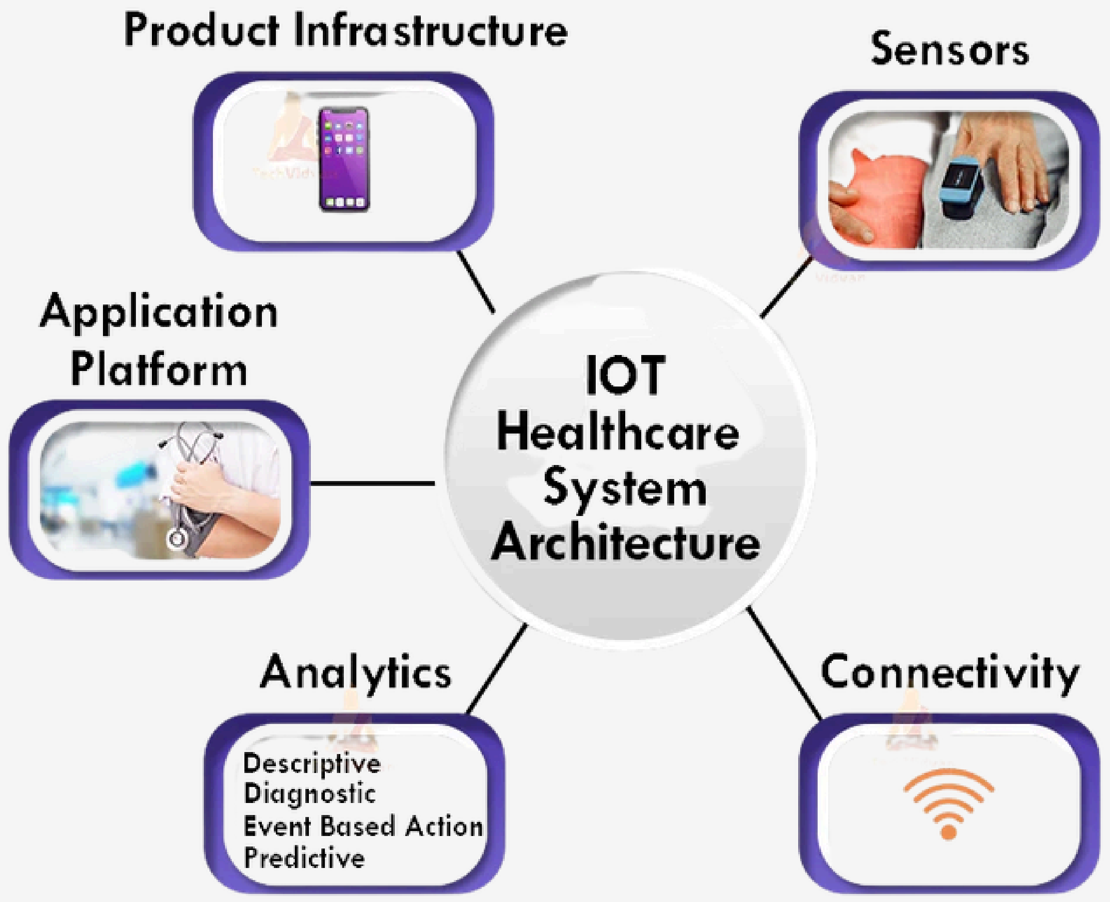

Fig 2 : IOT Healthcare System

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Problem Statement

- Lack of patient-level personalization
- Fragmented multimodal healthcare data
- Privacy risks in centralized learning
- Limited scalability of existing systems

Motivation And Objectives

Motivation

- ✓ Need for continuous healthcare monitoring
- ✓ Increasing privacy regulations
- ✓ Demand for secure data sharing
- ✓ Need for scalable healthcare intelligence

Objectives

- ✓ To design a federated digital twin framework for healthcare
- ✓ To integrate multimodal vital sign data
- ✓ To preserve privacy using federated learning
- ✓ To enable real-time health risk prediction
- ✓ To validate system performance experimentally

Literature Review

Literature Review

Existing studies show that digital twin technology is effective for personalized healthcare monitoring.

Federated learning and multimodal data analysis are also widely used to improve prediction accuracy while preserving data privacy.

Key Points:

- Digital twins enable patient-centric health modeling
- Federated learning supports secure, privacy-preserving analytics

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Related Studies (1/3)

A Revolution of Personalized Healthcare: Enabling Human Digital Twin With Mobile AIGC[1]

Contributions

- Introduced a holistic architecture for mobile AIGC-powered digital twins in healthcare
- Outlined lifecycle stages: data collection, management, training, and inference
- Explored real-world use cases, including virtual physical therapy
- Provided insights and open questions to inspire future research

Limitations

- Data Silos: Fragmented medical data limits collaboration; secure sharing is needed.
- Cross-Modal Complexity: Difficult to align text, images, and signals effectively.
- Model Efficiency: Large models increase energy and bandwidth costs; lightweight solutions are essential.

Related Studies (2/3)

Digital Twins for Cyber-Physical Healthcare Systems.[2]

Contributions

- Introduced core concepts of Digital Twins in healthcare
- Proposed reference architecture for cyber-physical healthcare systems
- Explored DT applications in precision medicine and patient modeling
- Reviewed current trends and challenges in healthcare DT research

Limitations

- Lack of real-time data processing in clinical settings
- Weak interoperability across healthcare systems
- Challenges in maintaining data privacy and quality
- Limited personalization in treatment modeling
- High computational cost for dynamic digital twins

Related Studies (3/3)

A digital twin framework for real-time healthcare monitoring: leveraging AI and secure systems for enhanced patient outcomes[3]

Contributions	Limitations
• Real-time monitoring using AI and sensor data • MXBoost model for accurate health predictions • MDT framework with cloud, IoT, and security layers • Autocorrelation analysis for long-term trends • Rolling average smoothing for stable forecasts	• No federated learning; only central training. • Personalization limited to body data. • Requires internet; not for offline or rural areas. • Uses basic RSA encryption; no advanced privacy. • Tested on small group (20 people) and one dataset.

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Research Gap

- Existing digital twin models are mostly centralized and generic.
- Multimodal healthcare data are not effectively fused in current systems.
- Privacy-preserving learning is applied without personalization.
- A complete federated digital twin framework for real-time healthcare monitoring is still missing.

Methodology

Methodology

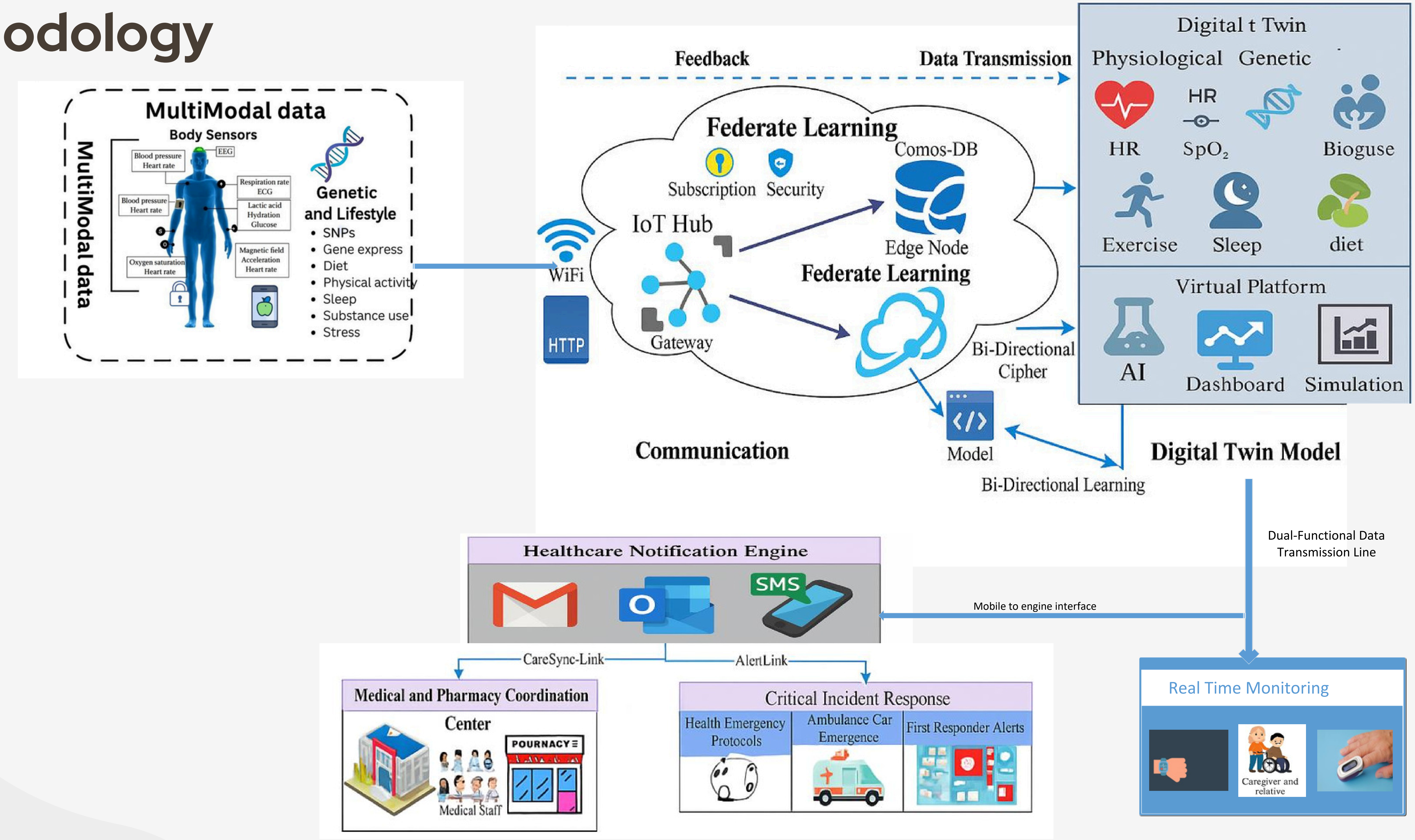

Fig 3 : Proposed architecture of the federated digital twin framework

Approach

- Local data are processed at edge devices to build patient digital twins.
- Federated learning aggregates model parameters without sharing raw data.
- The global model updates personalized digital twins continuously.

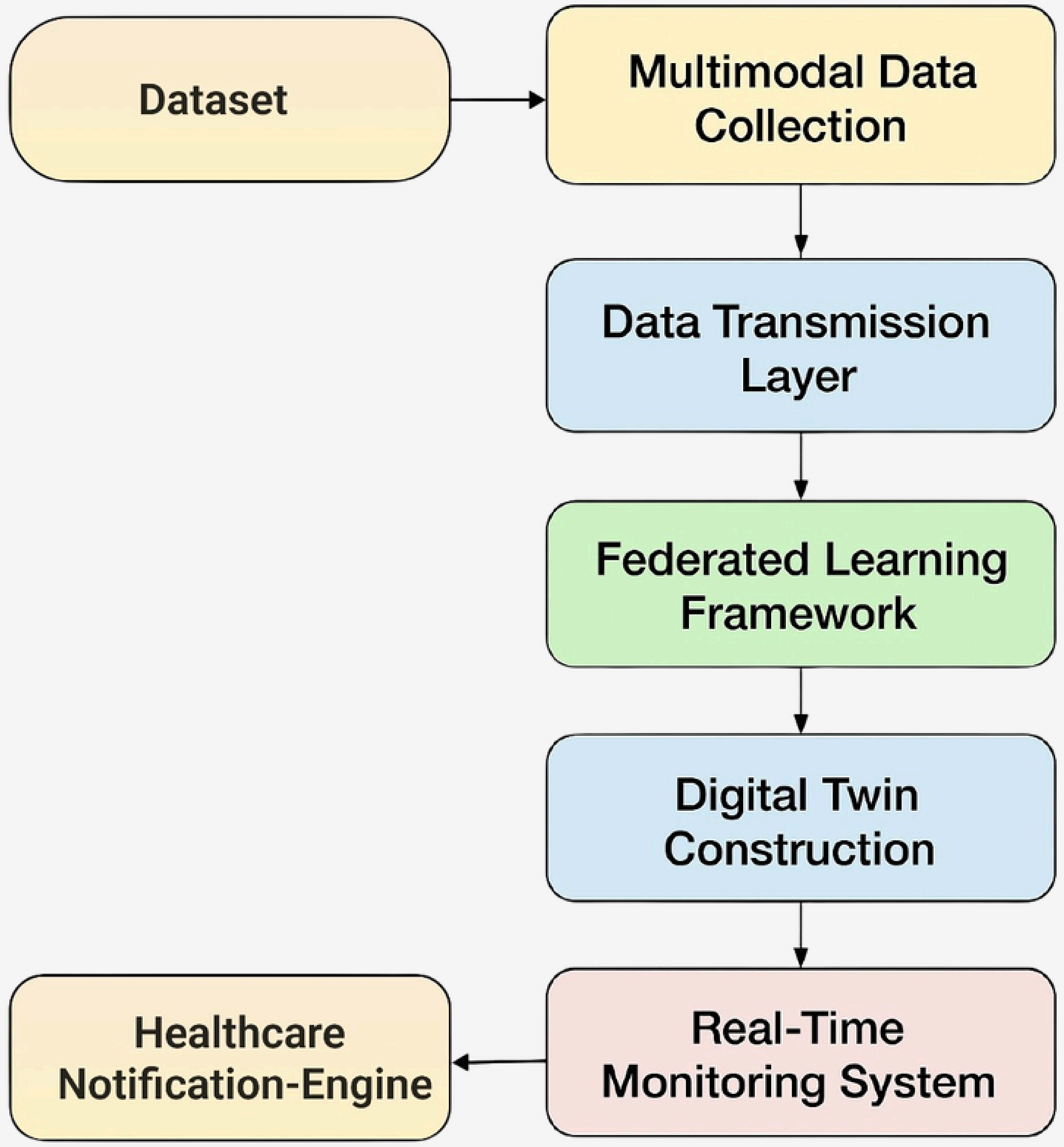

Fig 4 : Data Flow Diagram

Research Design

This study follows a quantitative and experimental research design.

The design is suitable for evaluating prediction accuracy and system performance.

Aspect	Description
Research Type	Quantitative
Approach	Experimental
Design Strategy	Model-based analysis
Data Source	Primary dataset (Kaggle)
Evaluation	Accuracy, Precision, Recall, F1-score, AUC

Table 3.1: Summary of Research Design

Data Collection Methods

Data Collection

Dataset Source: Kaggle – Human Vital Signs Dataset
Data Type: Heart rate, body temperature, oxygen saturation, and related vital indicators
Collection Method: Primary dataset
Sample Size: Multiple patient records representing realistic healthcare scenarios

Dataset Description

The Human Vital Signs Dataset contains real-world physiological health indicators collected from multiple individuals.It supports health risk classification and prediction tasks. The dataset is suitable for evaluating personalized healthcare monitoring models.

Data Analysis Methods

Step	Technique	Purpose
Data Cleaning	Missing value handling	Improve data quality
Normalization	Min–Max / Standard scaling	Ensure feature consistency
Encoding	Label encoding	Convert categorical data
Feature Selection	Correlation analysis	Select relevant features
Model Training	ML classifiers	Train prediction models
Validation	Train–test split	Evaluate model generalization
Evaluation	Accuracy, Precision, Recall, F1-score, AUC	Measure performance

Table 3.2: Data Analysis Methods

Gantt Chart

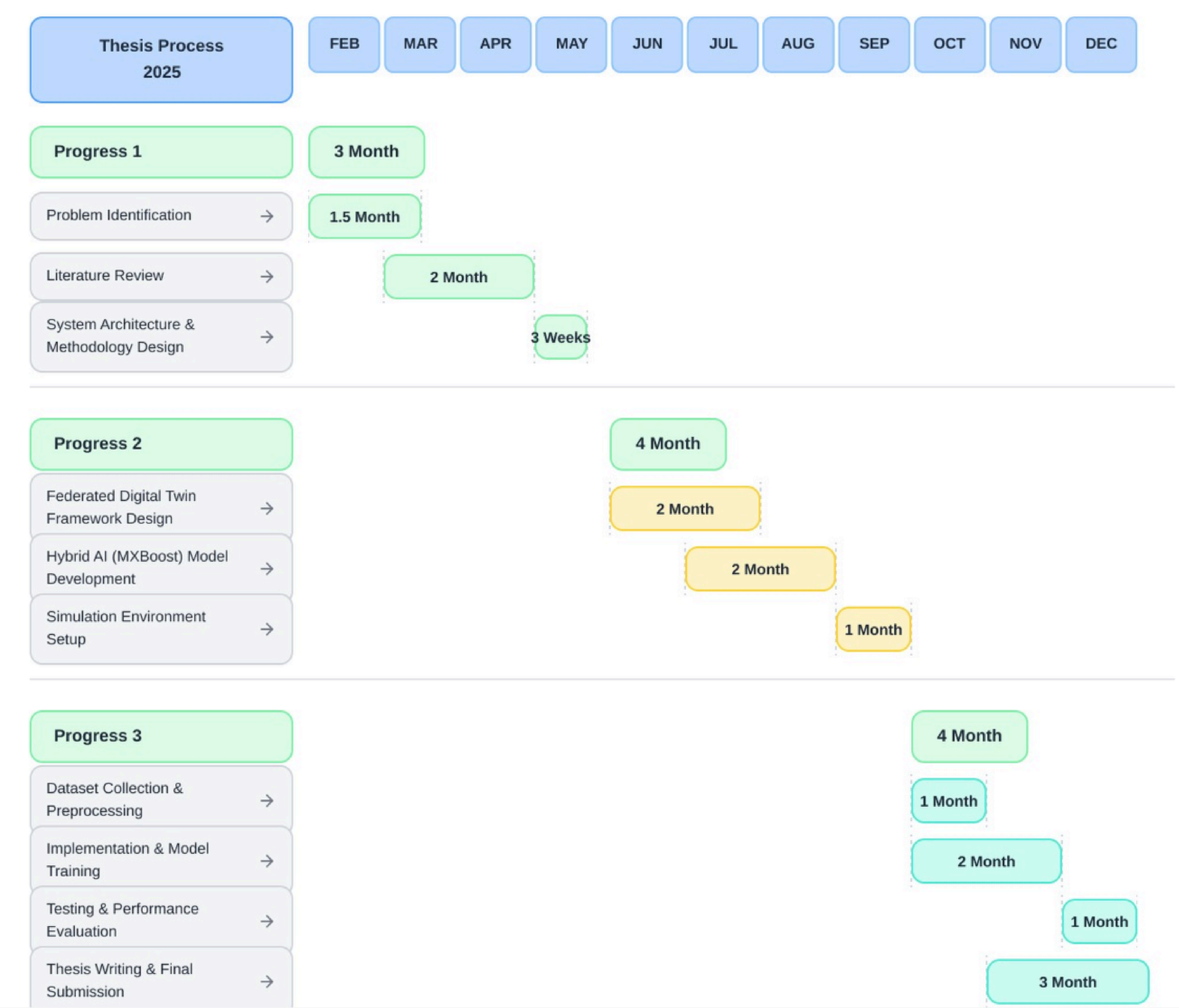

Fig 4 : Gantt Chart

Implementation

Patient Data

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Preprocessing & Scaling

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Split into K Clients

↓

Client 1 (MLP)

Client 2 (MLP) → FedAvg → Global MLP

Client K (MLP)

↓

Global Digital Twin Prediction

↓

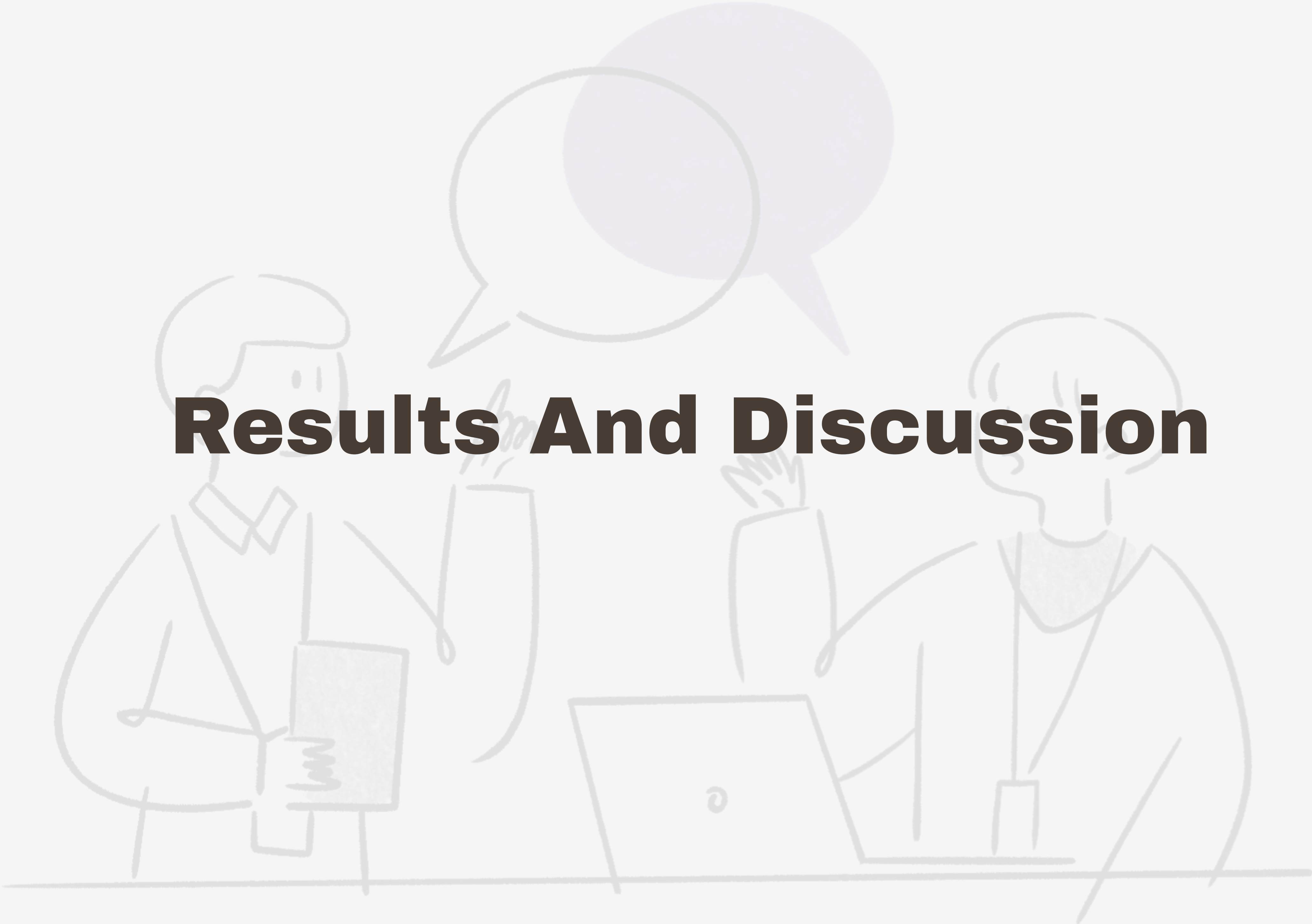
MXBoost Ensemble (MLP + XGBoost)

↓

Final Healthcare Decision

Fig 5 : Implementation work flow diagram

Results And Discussion



Results & Discussion

This section presents the experimental results and discussion of the proposed federated digital twin healthcare framework.

The system was evaluated using real patient vital sign data under a simulated federated learning environment.

Key findings, comparisons, and performance analyses are discussed using visual results.

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Key Finding

✓ Experiments were conducted in a simulated federated learning environment.

✓ Multimodal vital sign data improved prediction accuracy.

✓ Hybrid AI model (MXBoost) achieved the best performance.

✓ The system successfully detects abnormal health conditions.

Confusion Matrix

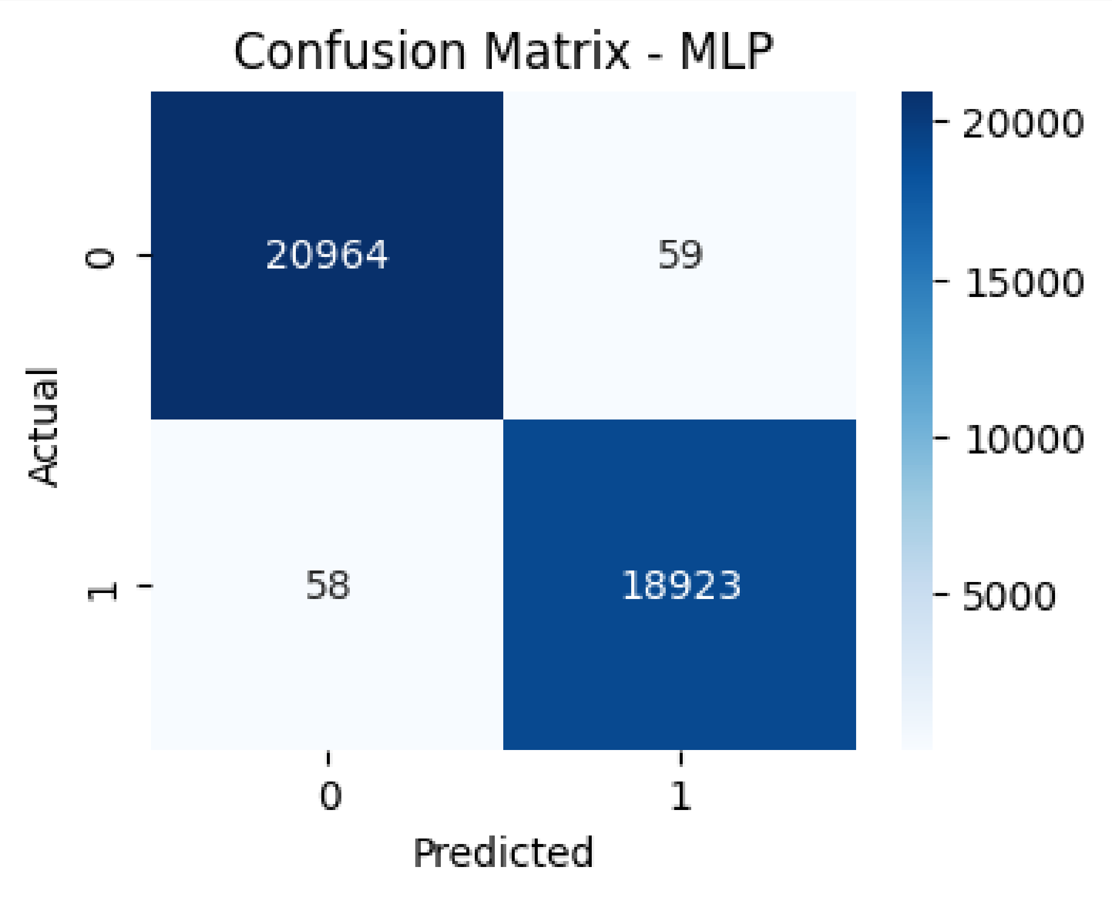

Fig 6: MLP: [[20964, 59], [58 , 18923]]

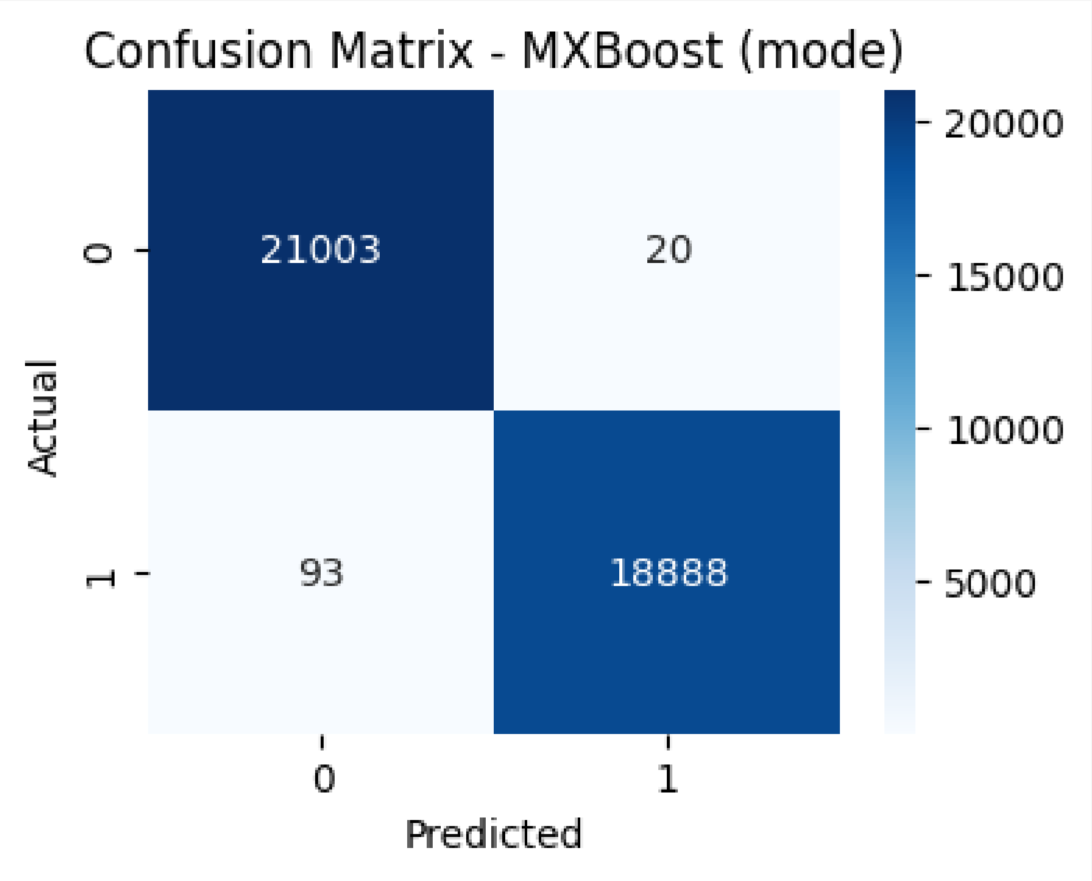

Fig 8: MXBoost: [[21003 , 20], [93 ,18888]]

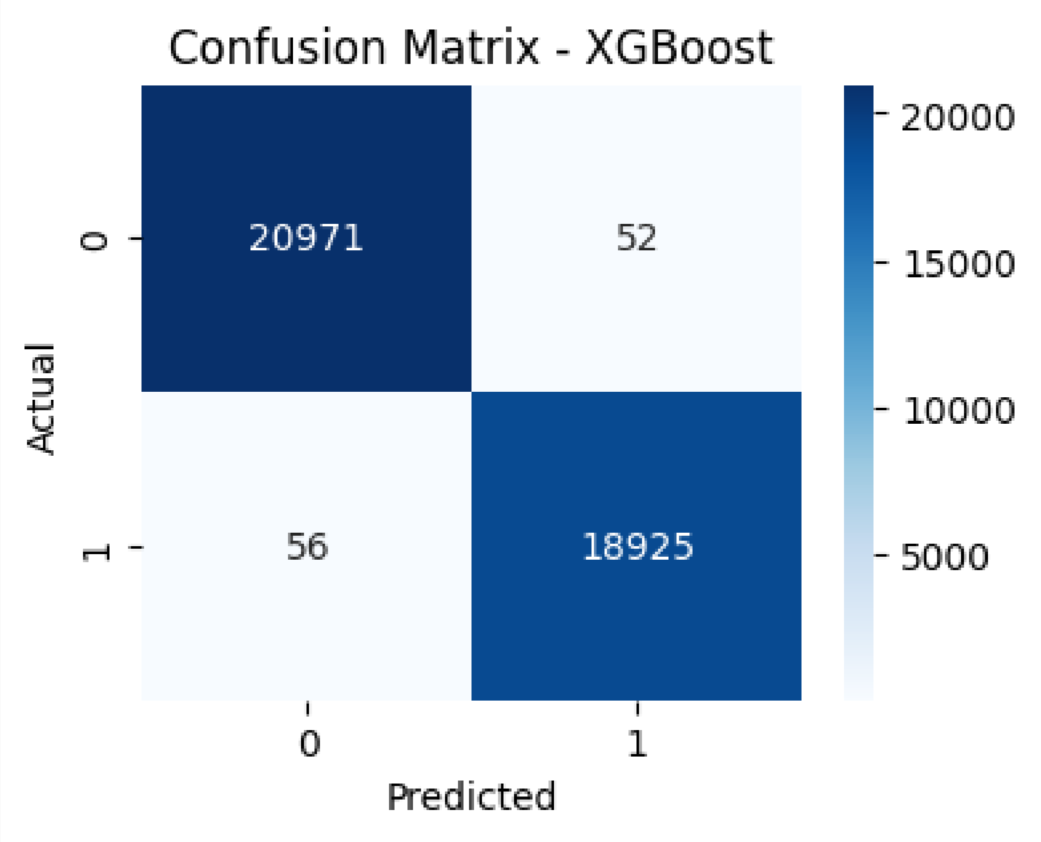

Fig 7: XGBoost: [20971 , 52], [56 , 18925]]

Overall performance

Model	Accuracy	Precision	Recall	F1-Score	AUC
MLP	0.9918	0.9935	0.9892	0.9913	0.9987
XGBoost	0.9946	0.9968	0.9929	0.9948	0.9992
MXBoost	**0.9972**	**0.9989**	**0.9951**	**0.997**	**0.9999**

Table 4.1 indicates that all models achieved accuracy above 99%. MXBoost shows the best overall performance with the highest accuracy, F1-score, and AUC. XGBoost and MLP also provide consistent and reliable prediction results.

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Federated learning and model aggregations.

Model	Accuracy	Precision	Recall	F1-Score
Local Models (Average)	0.9965	0.9965	0.9964	0.9965
Global Model (Final)	**0.997**	**0.9969**	**0.997**	**0.9969**

Table 4.2 presents the average local models and the global final updates with the 10 round of epoch and update the global models through the local one. This ensure the federated learning and model aggregations.

Comparison

Model / System	Accuracy (%)	F1-score	Precision	Response Time
Our Work (Federated Digital Twin + MXBoost)	99.7	0.9969	0.9969	6 ms
Human Digital Twin with Mobile AIGC [1]	95.26	0.95	0.95	94.53 s
Cyber-Physical Healthcare DT [2]	96	0.95	0.95	N/A
Real-Time Healthcare DT (AI-based) [3]	95.09	0.95	0.95	Minimal Delay
Clustered Federated Learning (CFL-IDS) [4]	81.9	0.77	0.82	0.25 s
Split Federated Learning for IoT [6]	82	0.82	0.82	N/A

Table 4.3: Performance Comparison with Existing Systems

Performance Analysis

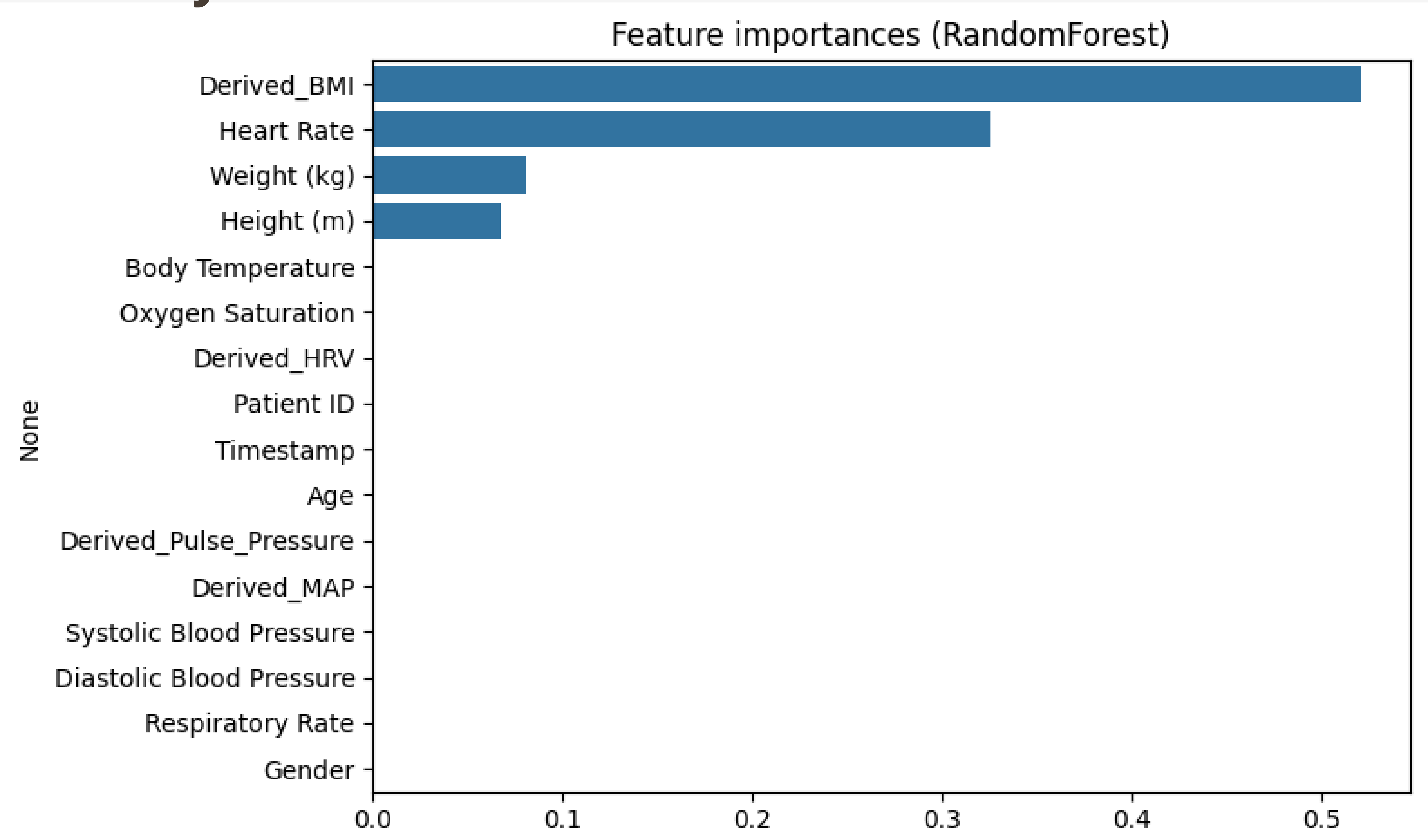

Fig 9: Horizontal bar chart of feature importance from Random Forest

Performance Analysis

Performance was evaluated using accuracy, precision, recall, and F1-score.
The hybrid MXBoost model consistently outperformed baseline models.
The results indicate reliable and stable healthcare risk prediction.

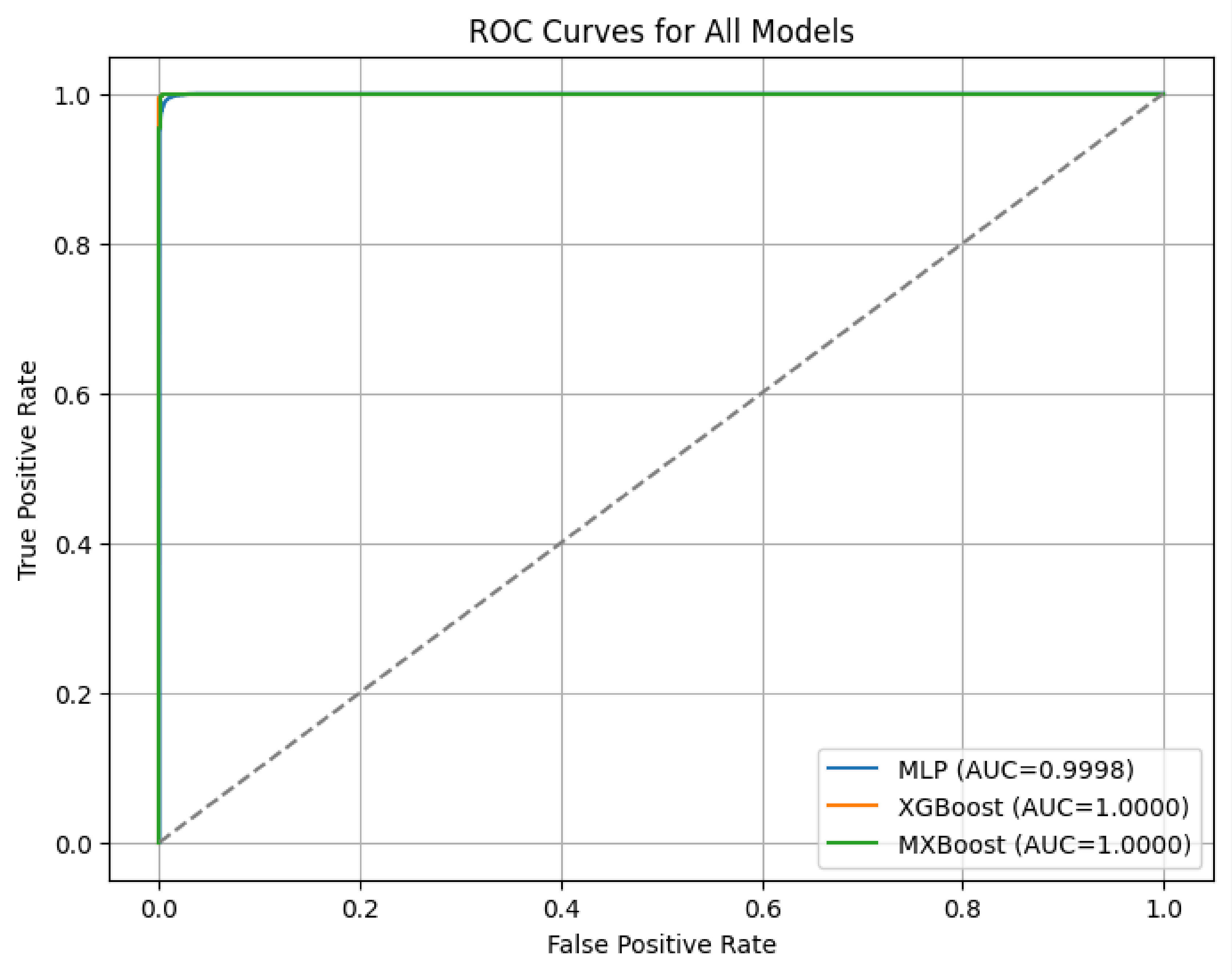

Fig 10: ROC Curves

Performance Analysis

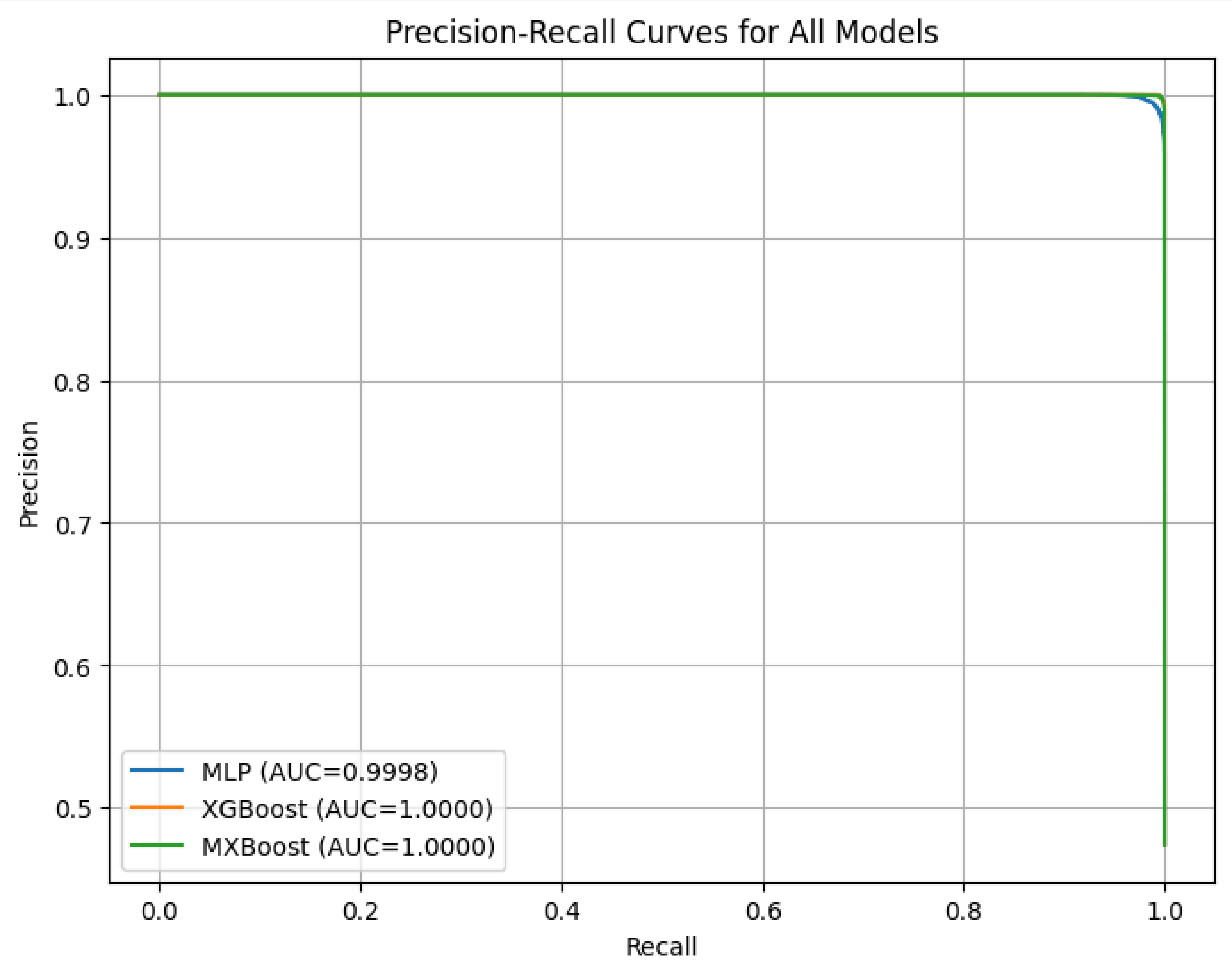

Fig 11: Precision-Recall Curve

Digital Twin Simulation

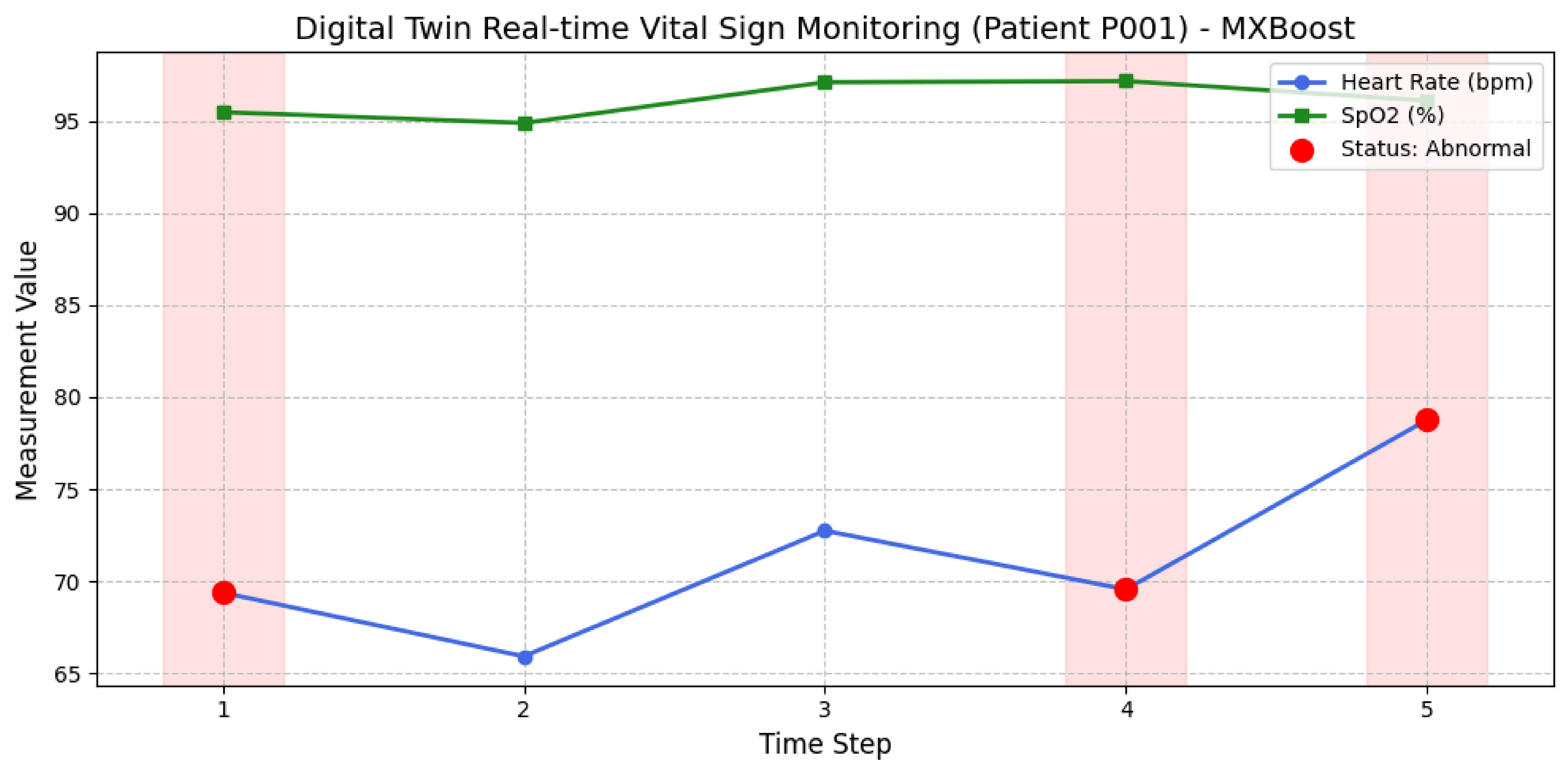

Fig 12: Horizontal bar chart of feature importance from Random Forest Real-time Digital Twin simulation for Patient P001

Digital Twin Simulation

Time	HR	SpO_2	BT	Status
1	69.38	95.48	36.93	Alert(Abnormal)
2	65.92	94.88	37.39	Normal
3	72.74	97.1	36.23	Normal
4	69.56	97.17	36.34	Alert(Abnormal)
5	78.76	96.1	36.68	Alert(Abnormal)

Table 4.4: Digital Twin Abnormal Reading Simulation Results

Conclusion

Summary of Key Findings

- The proposed federated digital twin framework successfully enables personalized healthcare monitoring.
- Integration of multimodal vital sign and lifestyle data improves health risk prediction accuracy.
- Federated learning ensures patient data privacy by eliminating raw data sharing.
- The hybrid MXBoost model achieves high predictive performance with reliable anomaly detection.
- Experimental results demonstrate the scalability and effectiveness of the proposed system for healthcare applications.

Contributions

✓ **Framework Design:** Proposed a secure and personalized federated digital twin framework for healthcare monitoring.

✓ **Multimodal Integration**: Integrated vital sign and lifestyle data to enable patient-centric health analysis.

✓ **Privacy Preservation**: Applied federated learning to ensure data privacy without sharing raw patient data.

✓ **Hybrid AI Model**: Developed a hybrid MXBoost model to achieve accurate health risk prediction.

✓ **System Evaluation**: Validated the proposed system using real patient data and standard machine learning metrics.

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Limitations and Future Work

Limitations

✓ The system was evaluated using limited publicly available healthcare datasets.

✓ Real-time deployment with live IoT sensor streams was not fully implemented.

✓ The federated learning setup was simulated, not tested across multiple real institutions.

✓ Explainability of hybrid AI (MXBoost) decisions is still limited for clinical interpretation.

✓ Communication and computational overhead in large-scale federated environments were not extensively analyzed.

Limitations and Future Work

Future Work

✓ Apply the proposed framework to real-time IoT and wearable sensor data.

✓ Validate the system using large-scale clinical datasets across multiple healthcare institutions
.

✓ Integrate deep learning models such as LSTM or Transformers for temporal health analysis.

✓ Enhance model explainability using AI techniques like SHAP or LIME.

✓ Optimize federated learning for lower latency, better scalability, and stronger privacy guarantees

Conclusion

This thesis presents a secure and personalized digital twin–based healthcare monitoring framework using IoT and hybrid AI techniques.

The proposed system addresses key challenges such as data privacy, limited personalization, and fragmented healthcare data through federated learning and multimodal data integration.

Experimental evaluation demonstrates that the hybrid MXBoost model achieves high prediction accuracy while enabling real-time health monitoring and early anomaly detection.

Overall, the proposed framework shows strong potential for scalable and privacy-preserving healthcare applications.

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THANK
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